# A FAILURE OF BASE CHANGE IS WITNESSED BY AN ALGEBRAICALLY CLOSED FIELD OVER A NORMAL AFFINE BASE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete reduction: the five steps, including the passage to constant cyclic coefficients, are written out and each reduction names the result that licenses it. The author's paragraph at source lines 17319-17323 stating the converse is outside the counted proof span and is not needed for the direction proved here.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

*Editorial note.* The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

*Allowed standard prerequisites (editorial).*

- The base change property BC(f, n, q). Same chapter, Remark remark-base-change-holds, recorded as the context of this card. Supplies the property whose failure is witnessed.
- Preparation of a diagram where base change fails. Same chapter, Lemma lemma-base-change-does-not-hold-pre. Supplies the starting commutative diagram and the sheaf for which the base change map fails.
- Base change through an integral top morphism, and through an integral bottom morphism. Same chapter, Lemmas lemma-base-change-q-integral-top and lemma-base-change-q-integral-bottom. Allow the base to be replaced by the normalization in the algebraically closed field without losing the failure.

*Source reference.* Original lemma 0F07: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 17079–17079. Navigation pointer invoked at proof line 17327; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Date*: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0F04: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16939–16939. Navigation pointer invoked at proof line 17330; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0F05: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16997–16997. Navigation pointer invoked at proof line 17336; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Remark* 0.1. Let $f : X \to S$ be a morphism of schemes. Let $n$ be an integer. We will say $BC(f, n, q_0)$ is true if for every commutative diagram

$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow[h]{} & Y \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle e} \\ S & \longleftarrow & S' & \xleftarrow{g} & T \end{array}$$

with $X' = X \times_S S'$ and $Y = X' \times_{S'} T$ and $g$ quasi-compact and quasi-separated, and every abelian sheaf $\mathcal{F}$ on $T_{étale}$ annihilated by $n$ the base change map

$$(f')^{-1} R^q g_* \mathcal{F} \longrightarrow R^q h_* e^{-1} \mathcal{F}$$

is an isomorphism for $q \leq q_0$.

**Lemma 0.2.** *With $f : X \to S$ and $n$ as in Remark 0.1 assume for some $q \geq 1$ we have that $BC(f, n, q-1)$ is true, but $BC(f, n, q)$ is not. Then there exist a commutative diagram*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow[h]{} & Y \\ {\scriptstyle f}\downarrow & & \downarrow & & \downarrow \\ S & \longleftarrow & S' & \longleftarrow & \mathrm{Spec}(K) \end{array}$$

*with both squares cartesian, where*

1. *$S'$ is affine, integral, and normal with algebraically closed function field,*
2. *$K$ is algebraically closed and $\mathrm{Spec}(K) \to S'$ is dominant (in other words $K$ is an extension of the function field of $S'$)*

*and there exists an integer $d|n$ such that $R^q h_*(\mathbf{Z}/d\mathbf{Z})$ is nonzero.*

*Proof.* First choose a diagram and $\mathcal{F}$ as in Lemma 0F07. We may and do assume $S'$ is affine (this is obvious, but see proof of the lemma in case of doubt). By Lemma 0F04 we may assume $K$ is algebraically closed. Then $\mathcal{F}$ corresponds to a $\mathbf{Z}/n\mathbf{Z}$-module. Such a modules is a direct sum of copies of $\mathbf{Z}/d\mathbf{Z}$ for varying $d|n$ hence we may assume $\mathcal{F}$ is constant with value $\mathbf{Z}/d\mathbf{Z}$. By Lemma 0F05 we may replace $S'$ by the normalization of $S'$ in $\mathrm{Spec}(K)$ which finishes the proof. □